Senior QA Home Assignment: Manual QA

Abiodun Oyawale

A1 - Requirements Review

Questions for PM / Designer (by impact)

1. Expired message: PRD says "This code has expired." Figma says "Promo code expired". Which is correct?
2. Figma shows a "Free shipping" badge, but the PRD says shipping is never discounted. Is there a free-shipping code type?
3. Is the minimum order value checked on the full subtotal or only on non-sale items?
4. Is the discount applied to all items or only non-sale items? What is the rounding rule?
5. A fixed discount larger than the eligible items: is it capped at the eligible amount?
6. Is tax calculated on the discounted amount? Client or backend?
7. Lockout: per user, device or session? Does success reset the count? Does it survive logout or restart?
8. Which failures count as attempts (invalid, expired, below minimum)?
9. What does "session" mean? What if the cart changes or the code expires before payment?
10. Does a code applied on web carry over to mobile?
11. Usage limits: per user, global cap, stacking with other offers?
12. Is whitespace trimmed? Is the field auto-uppercased?

Contradictions: PRD vs Design

PRD	Design
"This code has expired."	"Promo code expired"
Shipping never discounted	"Free shipping" badge shown on apply
Web, iOS and Android identical	Collapsed on mobile, expanded on web
No design for the "Add $X more" error	-

Gaps

- Replacing a code, and whether Remove resets failed attempts.
- Cart changes after a code is applied (falls below minimum, sale item added).
- Code expires between apply and payment.
- Rounding, and a fixed discount larger than the order.
- Missing messages: invalid, wrong format, locked, sale-only cart, network error.
- Lockout display (timer, message) and whether it persists across devices.
- Guest vs logged-in users.
- Offline or timeout behaviour on Apply.

Top 3 Risks

1. **Wrong totals.** Discount, tax, shipping, sale items, minimum and rounding interact. Errors mean lost revenue or overcharged customers.
2. **Platform drift.** Three separate clients can behave differently. If lockout is client-side, it can be bypassed, and codes can be brute-forced.
3. **Stale state.** A saved code can become invalid (expired, cart below minimum) and still be used if not rechecked at payment.

A5 - Mobile and Design Validation

Device matrix. Based on app analytics (top devices by usage); if unavailable:

	iOS	Android
Main	iPhone 15/16, iPhone SE	Galaxy S24, Pixel 8
Secondary	iPhone 12 (oldest supported OS)	Redmi/Xiaomi, Galaxy A-series (low-end)

Cover oldest and latest OS, small and large screens, stock and skinned Android.

Mobile risks

- Keyboard covering the field or button; Go/Done key; autocorrect changing the code; paste with spaces.
- Calls, notifications, backgrounding or app kill during Apply. Are the code and lockout timer kept?
- Rotation and font scaling: text, error and expanded state kept.
- Slow or lost network, double tap on Apply.
- VoiceOver/TalkBack, large text, contrast, touch target size.

Figma check. Compare spacing, font size, colours (#1A73E8, #D93025), states (empty, error, applied, locked) and copy. Tools: Figma Dev Mode, Xcode and Android Studio inspectors, side-by-side screenshots. Report each difference with both screenshots, measured vs expected values, device and OS. A state that was never designed is a question for the designer, not a bug.

A6 - Prioritization

1. **Beta first (10:00-10:30).** Possible payment loss. I only triage: reply to the client, ask for transaction IDs, times and error text, check logs and the payment provider, and hand findings to dev. Escalate if failures are ongoing.
2. **Alpha next.** Fixed 17:00 deadline. Run the remaining 60% of regression by risk (payments and core flows first). At 13:00 I check progress and warn the PM early if sign-off is at risk.
3. **Gamma last (after 16:00).** Short PRD review with the top questions and risks, due before 09:00.

Communication: client gets an acknowledgement and hourly updates; PM and dev lead get status on Alpha and Beta; the Gamma PM gets a heads-up that questions arrive tomorrow morning.

Not doing: root-causing Beta myself, writing full test cases for Gamma, or cutting Alpha's regression scope without telling anyone.

A7 - SQL

```
1. discount larger than subtotal
select id, user_id, subtotal, discount_amount
from orders
where discount_amount > subtotal;

2. code used after it expired
select o.id as order_id, pc.code, pc.expires_at, op.applied_at
from order_promotions op
join promo_codes pc on pc.id = op.promo_code_id
join orders o on o.id = op.order_id
where op.applied_at > pc.expires_at;

3. more than one code on an order
select order_id, count(*) as codes_applied
from order_promotions
group by order_id
having count(*) > 1;

4. total does not match the formula
select id, subtotal, discount_amount, tax, shipping, total,
       round(subtotal - discount_amount + tax + shipping, 2) as expected_total
from orders
where round(total, 2) <> round(subtotal - discount_amount + tax + shipping, 2);
```